

Mahin Shahriar

PhD Student

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EDUCATION

- **New York University (NYU), Tandon School of Engineering** Sep'25 - Ongoing
PhD. in Electrical and Computer Engineering (ECE) Brooklyn, New York, USA
– CGPA: **3.889/4.00** (First Semester)
- **Bangladesh University of Engineering and Technology (BUET)** Apr'19 - Jul'24
BSc. in Electrical and Electronic Engineering (EEE) Dhaka, Bangladesh
– CGPA: **3.72/4.00** (Major Electrical Energy and Power Systems (EEPS) **3.86**)
- **Notre Dame College (NDC)** Jul'16 - Jul'18
Higher Secondary Certificate Dhaka, Bangladesh
– GPA: **5.00/5.00**
– General scholarship recipient

RESEARCH INTEREST

- Application of Artificial Intelligence in Power System
- Smart Grid
- Power System modeling and simulation
- Micro Grid
- Power System Monitoring
- Power System Data Analytics
- Power System Sensor Placement

RELEVANT COURSEWORK

Energy Conversion I & II
Power Electronics
Power System Protection
High Voltage Engineering

Power System I & II
Power System Operation and Control
Power Transmission and Distribution
Smart Grid

RESEARCH EXPERIENCE

- **Graph-Learning-Based High-Granularity Voltage Visibility in Low-Voltage Networks With Strategic Metering** Sep'25-Ongoing
This research proposes a data-driven framework to achieve high-resolution customer-voltage visibility in LV networks via a sparse deployment of HRRSMs alongside conventional SM data. It constructs a mutual-information-based correlation graph and applies an edge-weighted GraphSAGE model to infer minute-level voltage profiles at unobserved nodes. The results demonstrate accurate three-phase voltage reconstruction without requiring dense high-rate metering.
- **High-Accuracy Power Quality Disturbance Classification Using Vision Transformers and GASF Encoded Signals** Aug'24-Jul'25
The research involves classifying power quality disturbance signals encoded as GASF images. It also demonstrates the effectiveness of Vision Transformers compared to Convolutional Neural Networks in capturing signal patterns for real-time power system diagnostics and reliability enhancement.
- **Detection & Classification of Transmission Line Faults Using Convolutional Neural Networks and Vision Transformer (UG Thesis)** Jul'23-Jun'24
The thesis focuses on creating custom datasets for complex power system networks using efficient algorithms within Simulink models. It leverages Convolutional Neural Network architectures and computer vision techniques to improve performance and accuracy.

PUBLICATIONS

1. Shahriar, M., Ghosh, M., & Chowdhury, A. H. (2025). Detection and classification of transmission line faults using convolutional neural networks. In *2025 international conference on electrical, computer and communication engineering (ecce)* (p. 1-6). doi: [10.1109/ECCE64574.2025.11013912](https://doi.org/10.1109/ECCE64574.2025.11013912)
2. Shahriar, M., Ali, M., & Chowdhury, A. H. (2025). Comparative load flow analysis using artificial neural network and the newton-raphson method. In *2025 international conference on electrical, computer and communication engineering (ecce)* (p. 1-6). doi: [10.1109/ECCE64574.2025.11014048](https://doi.org/10.1109/ECCE64574.2025.11014048)
3. Asad, D. M., Shahriar, M., & Chowdhury, A. H. (2025). Load forecasting for dhaka city using rnn, lstm, and gru architectures with meteorological and temporal data. In *2025 international conference on electrical, computer and communication engineering (ecce)* (p. 1-6). doi: [10.1109/ECCE64574.2025.11013779](https://doi.org/10.1109/ECCE64574.2025.11013779)

PROFESSIONAL EXPERIENCE

- **Academic Internship:** Completed a **thirteen-day** academic internship program organized by the **Dhaka Electric Supply Company Limited (DESCO)**.
- **Industrial Visit:** Completed site visit at **National Load Dispatch Center (NLDC)** and **Rooppur Nuclear Power Plant**.

NOTABLE PROJECTS

- **Power Flow Analysis of IEEE 5 Bus using Artificial Neural Network** (Course Project) (Nov'23–Jun'24)

Power flow analysis was performed on an IEEE five-bus system using an **ANN** architecture overcoming the limitations of traditional methods like the Newton Raphson.

 - * **Project Link:** https://github.com/MahinShahriar1/Power_Project-on-ANN
 - * **Tools & Technology:** MATLAB, Python
- **Electric Service Design** (Course Project) (Nov'23–Jun'24)

This project focuses on developing a precise electrical plan to ensure tenant comfort and safety while converting a residential property into a ten-story complex, incorporating modern features such as a garage and a rooftop pool.

 - * **Project Link:** <https://github.com/MahinShahriar1/Electric-Service-Design-Project>
 - * **Tools & Technology:** AutoCAD
- **One Hour POMORDO Clock** (Course Project) (Nov'22–May'23)

This project emphasizes the fabrication of a digital clock capable of performing various clock mechanisms, including a stopwatch and timer.

 - * **Project Link:** https://github.com/MahinShahriar1/Digital_Electronics_Project
 - * **Tools & Technology:** Proteus
- **Epileptic Seizure Detection using EEG Signal Processing** (Course Project) (May'22–Nov'22)

This project focuses on implementing an **LDA** machine learning model to detect epileptic seizures.

 - * **Project Link:** https://github.com/MahinShahriar1/DSP_Project
 - * **Tools & Technology:** MATLAB
- **Advanced Home Automation System with WiFi & Mobile App Based Interface** (Course Project) (May'22–Nov'22)

This project utilized WiFi **ESP32/IoT** modules and the **Blynk** mobile app to establish wireless connectivity for controlling appliances such as lights, fans, and monitors. It also enabled the saving of real-time sensor data on a mobile device or the **Blynk** server.

 - * **Project Link:** <https://github.com/MahinShahriar1/Control-System-Project>
 - * **Tools & Technology:** IoT based electronics components, Arduino
- **Panzer Fight(Arcade Game)** (Course Project) (Feb'21–Nov'21)

The game was based on the Gauss-Jordan method with pivot. Two panzers fire shells at each other, with three walls placed between them as barriers.

 - * **Project Link:** https://github.com/MahinShahriar1/MATLAB_Project
 - * **Tools & Technology:** MATLAB

SKILLS

- **ML Frameworks and Libraries:** Deep Neural Network (using TensorFlow, Keras, PyTorch and Hugging Face), Image processing (using OpenCV)
- **Hardware Skills:** Arduino, FPGA, ATMGEA32
- **Languages:** English (Professional), Bengali (Native)
- **Performing Arts:** Public Speaking
- **Document Preparation:** Overleaf(LaTex), Microsoft Office, Microsoft Excel, Microsoft Word, Microsoft Power-point
- **Programming Languages:** C/C++, Python, System Verilog, MATLAB
- **Tools and Technology:** Simulink, Quartus, Pycharm, ModelSim, PSpice, Proteus, AutoCAD, PSAF, OpenDSS, PSSE

AWARDS & ACHIEVEMENTS

- EEE Faculty **Dean's List Award** from BUET.
- **Board scholarships** e.g., HSC, SSC.

REFERENCES

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